Timothy Bangma

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GitHub (LLM Assisted): github.com/sipak-gh · SourceHut (Human Only): git.sr.ht/~sipak

Summary

Backend software engineer with 9+ years of professional experience specialising in Kotlin, TypeScript, and Golang. Proven track record designing and delivering distributed systems and microservices architectures at scale, from greenfield cloud-native platforms to zero-downtime legacy migrations. Experienced in REST API design, system design, CI/CD pipeline ownership, and driving features from ideation through long-term maintenance. Active in cross-team technical planning and mentorship of junior engineers.

Technical Skills

Languages: Kotlin · Java · TypeScript · Python · Golang · JavaScript · Ruby · Rust · Zig
Frameworks: SpringBoot · Node.js · React · React Native
Infrastructure: Kubernetes · AWS · GCS · Docker · Terraform · Jenkins · Helm · Apache Dataflow
Databases: PostgreSQL · Cloud Spanner · Redis · MongoDB · Oracle SQL · ElasticSearch
Observability: Datadog · PagerDuty · Grafana
Tools: CloudFormation · Troposphere · Git · GitHub Copilot · Claude Code · Linux · GNU Toolchain

Work Experience

EA Maxis, Server Engineer

Nov 2021 – Present
`Kotlin · SpringBoot · TypeScript · Apache Dataflow · ElasticSearch · Kubernetes · GCS`

- Collaborate with game client teams to plan, scope, and design Expansion Pack features, applying system design principles around response times, caching strategies, and external service dependencies to produce robust server-side specifications
- Leading development of new unreleased gaming products using agentic AI-assisted workflows (Claude Code via GitHub Copilot CLI), primarily in TypeScript and JavaScript
- Designed and built custom agentic developer workflows: feedback loops integrating build, test, lint, and codegen steps that run continuously as the agent works, enabling autonomous iteration with minimal manual intervention
- Extended agent capabilities with LSP tooling integration, allowing the agent to detect and self-correct code errors across the codebase without excessive human oversight
- Drove a zero-downtime migration of a legacy C++ content-hosting service to a Kotlin/SpringBoot/Kubernetes architecture, eliminating multi-day maintenance windows and reducing deploy time to under one hour, while maintaining full backward compatibility with existing clients (18k+ DAU)
- Serve as primary maintainer on several projects and act as a go-to resource for problem-solving across the main codebase
- Deliver knowledge-sharing sessions on major features built in shared libraries consumed by multiple game teams
- Mentor a junior engineer directly and provide ongoing technical guidance to mid-level team members

Arctic Wolf, Software Developer, Infrastructure (Member of Technical Staff)

Nov 2020 – Oct 2021
`Golang · Python · CloudFormation · Troposphere · AWS · Kubernetes`

- Built high-throughput, real-time data processing applications within Arctic Wolf's security telemetry pipeline, ingesting and normalising security data from diverse sources, including system metrics, network logs, disk usage, and monitoring platforms (Grafana, Datadog), at scale for downstream security services
- Developed internal Python tooling and build utilities to improve developer productivity and streamline the team's development workflow

HP Inc, Software Developer

May 2018 – Oct 2020
`Java · SpringBoot · NodeJS · Redis · MongoDB · Docker · AWS · Terraform · Jenkins`

- Developed a cloud-native microservice platform (~10 services, team of 6–8) providing on-demand printing to customers across North America and China
- Progressed to **Lead DevOps Engineer**, taking full ownership of CI/CD pipeline stability, Jenkins infrastructure, Terraform environments in AWS, and security automation including certificate and key rotation

ArcelorMittal Dofasco, Software Developer

Jan 2016 – Apr 2018
`Java · Kotlin · SpringBoot · NodeJS · PL/SQL · Oracle SQL`

- Built an automated sales reporting service that extracted and formatted steel inventory data from Oracle databases into actionable reports, accelerating the sale of time-sensitive devalued inventory
- Joined as a co-op student and was offered a full-time position upon graduation

Notable Projects

- **sipak-gh.github.io** (JavaScript): My project portfolio, a dependency-free static site that renders each project's README live from GitHub. A GitHub Action refreshes the project metadata from the GitHub API.
- **courseware** (JavaScript): An open library of 16 long-form courses spanning C, assembly, Zig, Go, React, Git and philosophy. I write them to deepen my own expertise, since teaching a subject end to end is the best test of understanding it, and share them freely with the community through a book-style reader rendered live from each course repo.
- **zen** (Zig): `ze`, a tiny modal editor following the influence of vi. Meant to get out of your way.
- **parc** (Go): A privacy-focused public archival tool, allowing you to encrypt files and filenames with your ed25519 SSH key. Built on age.
- **vibabel** (Rust): A fun terminal vi-based editor to let you modify the *nearly* infinite pages of Borges' Library of Babel.
- **esp32-firmware** (Xtensa Assembly): from-scratch course on bare-metal ESP32 firmware, including a 208-byte Xtensa assembly "Hello, World!" and a custom firmware-image packer verified against Espressif's tool.

Education

Mohawk College, Software Development Advanced Diploma (Co-op)
Sept 2014 – Dec 2017

Interests

Guitar · Amateur powerlifting · Low-level and embedded programming in Rust and Zig, including building custom firmware for ESP32 microcontrollers